

UK Visa Application Guidance & Document Readiness Agent

Problem Framing Document

1. Problem and Background

Applying for a UK visa can be a complicated process, especially for someone who is unfamiliar with the UK immigration system. The official information is available through GOV.UK, but applicants may need to read several pages and guidance documents to understand which requirements apply to their particular visa route.

For example, a person applying for a Skilled Worker visa may need to understand information about sponsorship, eligibility, English language requirements, financial requirements, supporting documents and dependant applications. The information is available, but the applicant still has to find the correct guidance, understand it and determine which parts are relevant to their own situation.

The problem this project addresses is therefore not a lack of information. The information already exists. The problem is the **time and effort required to find, interpret and connect the relevant official information**.

Problem Statement

Prospective UK visa applicants spend significant time searching and interpreting multiple sources of official guidance to understand the requirements and documents relevant to their visa application.

The proposed system aims to make this process easier by allowing users to ask questions in natural language and receive answers grounded in official UK visa guidance.

The system is designed as a **guidance and document-readiness assistant**, not as a replacement for GOV.UK, an immigration lawyer or a professional immigration adviser.

2. Primary User Persona

The primary user is a **prospective UK visa applicant**.

This could be someone considering a Skilled Worker, Student, Visitor or Family visa who wants to understand the basic requirements before starting or progressing with an application.

A typical user may not know exactly which GOV.UK page contains the information they need. They may also ask follow-up questions based on previous answers.

For example:

"What documents do I need for a Skilled Worker visa?"

followed by:

"What about my wife?"

The second question depends on the context of the first one. This makes conversation memory useful for the proposed system.

The agent could also support immigration consultants or visa caseworkers as a research and document-search assistant, but the main persona for this project is the prospective applicant.

3. Current Workflow and Pain Points

A typical applicant's current process can be represented as:

Question → Search GOV.UK → Open several pages → Read guidance → Identify relevant requirements → Work out required documents → Search again when a follow-up question arises

There are several problems with this process.

First, the applicant may not know the correct terminology to use when searching. Second, information can be spread across different guidance pages. Third, the applicant may find it difficult to determine which information is relevant to their specific situation.

This creates unnecessary search effort and increases the possibility that a user may misunderstand or overlook an important requirement.

The proposed agent addresses the search and information-understanding part of this process.

4. Inputs and Outputs

The main input is a natural-language question from the user.

Examples include:

- "What documents are required for a Skilled Worker visa?"
- "What financial evidence is needed for a Student visa?"
- "Can my wife apply as my dependant?"
- "I have a sponsor and passport. What else do I need?"
- "What about my wife?" as a follow-up question.

The agent uses the user's question together with the indexed GOV.UK guidance to generate a response.

The expected output is a clear, concise explanation based on the available source material. Where appropriate, the agent can identify relevant documents or requirements and explain what information it found.

If relevant information cannot be found, the agent should say so rather than inventing an answer.

5. Scope and Constraints

The system focuses on UK visa guidance and document readiness.

It should not:

- Guarantee that a user will receive a visa.
- Make a final immigration decision.
- Present itself as an immigration lawyer.
- Invent requirements when information is unavailable.
- Treat general model knowledge as a substitute for the indexed guidance.
- Act as a general-purpose assistant for unrelated questions.

The project also has a knowledge-freshness limitation. GOV.UK guidance can change, so the documents used by the RAG system need to be refreshed when official guidance changes.

6. Success Criteria

The project is considered successful if the agent can:

1. Retrieve relevant information from the indexed GOV.UK guidance.
2. Generate relevant and understandable answers.
3. Use tools appropriately for different types of questions.
4. Handle questions requiring more than one step.
5. Maintain context across multiple turns.
6. Adapt its response based on user feedback.
7. Avoid hallucinating when relevant information is unavailable.
8. Handle API and runtime failures without crashing.
9. Provide consistent results across repeated test questions.

The project therefore measures more than just whether the final answer "sounds good". It also considers retrieval, tool selection, memory, safety and failure handling.

7. Known Failure Cases and Edge Scenarios

Several failure cases were considered during development.

One is **missing information**. The relevant guidance may not be included in the knowledge base, in which case the agent should not attempt to create an answer from unsupported information.

Another is **outdated information**. If GOV.UK updates its guidance after the documents have been embedded, the agent could potentially return information that is no longer current.

The system may also receive ambiguous questions where there is not enough context to provide a reliable answer. In such cases, clarification is preferable to making assumptions.

Technical failures are another consideration. The Gemini API may reach a quota limit or become temporarily unavailable. The system therefore needs graceful failure handling rather than simply displaying a Python exception to the user.

Finally, there is a safety risk if users treat the agent's response as definitive legal or immigration advice. The system is therefore deliberately positioned as a guidance and document-readiness assistant.

8. Summary

The project addresses a practical information-search problem rather than simply creating a generic AI chatbot. The objective is to make official UK visa guidance easier to access and understand while keeping the user grounded in authoritative source material.

The use of RAG, tools, planning, memory and adaptive behaviour allows the agent to handle more realistic interactions than a single LLM question-and-answer system. However, the prototype remains limited by the freshness and coverage of its knowledge base and should not be treated as a replacement for official GOV.UK guidance or professional immigration advice.